ACM India Summer School on Symmetric Key Cryptography

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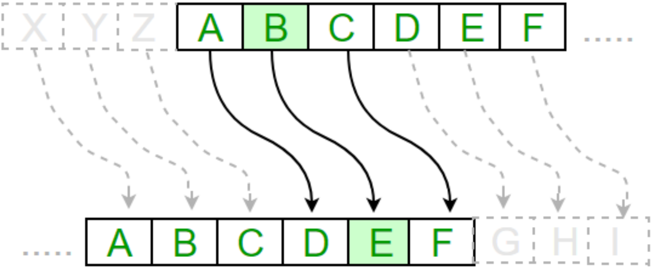
Practical Cryptanalysis and Programming Exercises

Outline

Caesar Cipher

Ciphertext

KHOOR ZRUOG



Caesar Cipher

Key: 3

Recovered Plaintext

HELLO WORLD

Method: Shift each letter three positions backward.

Caesar Cipher-Exercise

Ciphertext

Fdhvdu flskhu lv rqh ri wkh hduolhvw dqg vlpsohvw phwkrgv ri hqfubswlrq. Lw zrunv eb vkliwlqj hdfk ohwwhu lq wkh sodlqwhaw eB d ilahg qxpehu ri srvlwlrqv lq wkh doskdehw. Dowkrxjk lw lv hdvb wr lpsohphqw, lw surylghv yhub olwwoh vhfxulwb djdlqvw prghuq fubswdqdobvlv whfkqltxhv. Qhyhuwkhohvv, lw uhpdlqv d xvhixo hgxfdwlrqdo wrro iru xqghuvwdqglqj wkh edvlf frqfhswv ri fodvvlfdo fubswrjudskb.

Caesar Cipher-Exercise

Figure: Ciphertext

Caesar Cipher-Exercise

Key: 3

Recovered Plaintext

Caesar cipher is one of the earliest and simplest methods of encryption.

Caesar Cipher-Exercise

Key: 3

Recovered Plaintext

Caesar cipher is one of the earliest and simplest methods of encryption. It works by shifting each letter in the plaintext by a fixed number of positions in the alphabet. Although it is easy to implement, it provides very little security against modern cryptanalysis techniques. Nevertheless, it remains a useful educational tool for understanding the basic concepts of classical cryptography.

Shift Cipher

Ciphertext

YMJ VZNHP GWTBS KTC OZRUX TAJW YMJ QFED ITL

Shift Cipher

Ciphertext

YMJ VZNHP GWTBS KTC OZRUX TAJW YMJ QFED ITL

Key: 21

Recovered Plaintext

THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

Observation

- Caesar cipher has only 25 possible keys.
- Exhaustive search is practical.

Shift Cipher-Exercise

Ciphertext

Ynulpkcnwlddu eo w bqjzwiajpwh pkkh ej ejbkniwpekj oayqnepu. Ep dahlo Inkpayp iaoowcao bnki qjwqpdknevaz wyyaoo xu pnwjobkniejc lhwej pazp ejpk wj qjnawzwxha bkni. Whpdkqcd ikzanj yeldano wna iqyd ikna ykilhaz, deopkneywh oydaiao oqyd wo pda odebp yeldan naiwej qoabqh bkn hawnjejc pda xwoeyo kb ajynulpekj wjz ynulpwjwhuoeo.

Shift Cipher-Exercise

Figure: Ciphertext

Shift Cipher-Exercise

Key: 22

Recovered Plaintext

Cryptography is a fundamental tool in information security.

Shift Cipher-Exercise

Key: 22

Recovered Plaintext

Cryptography is a fundamental tool in information security. It helps protect messages from unauthorized access by transforming plain text into an unreadable form. Although modern ciphers are much more complex, historical schemes such as the shift cipher remain useful for learning the basics of encryption and cryptanalysis.

Atbash Cipher

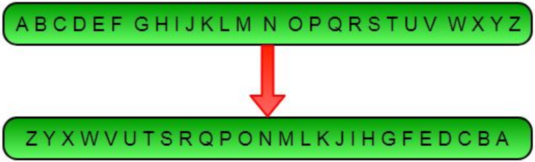
Ciphertext

GSRH RH Z HVXIVG NVHHZTV

Atbash Cipher

Ciphertext

GSRH RH Z HVXIVG NVHHZTV

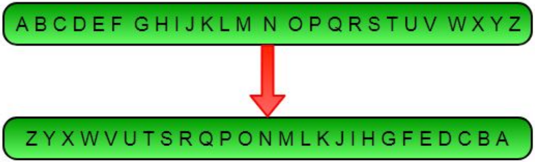


The key

Atbash Cipher

Ciphertext

GSRH RH Z HVXIVG NVHHZTV



The key

Recovered Plaintext

THIS IS A SECRET MESSAGE

Atbash Cipher-Exercise

Ciphertext

Xibkgltizksb rh gsv hxrvmxv lu hvxfirmt rmulinzgrlm yb gizmhulinrmt rg rmgl z ulin gszg rh wruurxfog gl fmwvihgzmw drgslfg gsv kilkvi pvb. Xozhhrxzo xrksvih hfxs zh gsv Zgyzhs xrksvi kilerwv z hrnkov rmgilwfxgrlm gl gsv xlmxvkgh lu vmxibkgrlm zmw wvxibkgrlm. Zogslfts rg luuvih orggov kizxgrxzo hvxfirgb, rg ivnzrmh zm rnkligzmg vwfxzgrlmzo gllo uli hgfwbrmt gsv srhglib zmw velofgrlm lu xibkgltizksrx gvxsmrjfvh.

Atbash Cipher-Exercise

Figure: Ciphertext

Atbash Cipher-Exercise

Recovered Plaintext

Cryptography is the science of securing information by transforming it into a form that is difficult to understand without the proper key.

Atbash Cipher-Exercise

Recovered Plaintext

Cryptography is the science of securing information by transforming it into a form that is difficult to understand without the proper key. Classical ciphers such as the Atbash cipher provide a simple introduction to the concepts of encryption and decryption. Although it offers little practical security, it remains an important educational tool for studying the history and evolution of cryptographic techniques.

Affine Cipher

Encryption

$$E(x) = (ax + b) \bmod 26$$

Decryption

$$D(y) = a^{-1}(y - b) \bmod 26$$

Affine Cipher

Plaintext

HELLO

$a = 5, \ b = 8$

Affine Cipher

Plaintext

HELLO

$a = 5, \ b = 8$

Plaintext	x	**Calculation**	**Cipher Value**	**Cipher**
H	7	$(5 \times 7 + 8) = 43 \bmod 26$	17	R

Affine Cipher

Plaintext

HELLO

$a = 5, \ b = 8$

Plaintext	x	Calculation	Cipher Value	Cipher
H	7	$(5 \times 7 + 8) = 43 \bmod 26$	17	R
E	4	$(5 \times 4 + 8) = 28 \bmod 26$	2	C
L	11	$(5 \times 11 + 8) = 63 \bmod 26$	11	L
L	11	$(5 \times 11 + 8) = 63 \bmod 26$	11	L
O	14	$(5 \times 14 + 8) = 78 \bmod 26$	0	A

Ciphertext

RCLLA

Affine Cipher-Exercise

Ciphertext

SPYFZAMPIFRY WU ZRC USWCVSC AH UCSEPWVM WVHAPQIZWAV NY ZPIVUHAPQWVM WZ WVZA I HAPQ ZRIZ WU XWHHWSELZ ZA EVXCPUZIVX OWZRAEZ ZRC FPAFCP GCY. SLIUUWSIL SWFRCPU UESR IU ZRC IHHWVC SWFRCP FPAJWXC I UWQFLC WVZPAXESZWAV ZA ZRC SAVSCFZU AH CVSPYFZWAV IVX XCSPYFZWAV.

Values

$$a = 5, b = 8$$

Affine Cipher-Exercise

Figure: Ciphertext

Affine Cipher-Exercise

Values

$$a = 5, b = 8,$$

$$a^{-1} = ?mod26.$$

Finding the Modular Inverse of $5 \pmod{26}$

We want to find an integer x such that:

$$5y \equiv 1 \pmod{26} \tag{1}$$

An inverse exists because $\gcd(5, 26) = 1$. We perform the division steps to find the Greatest Common Divisor and rewrite them:

$$\begin{aligned} 26 &= 5 \times 5 + 1 \\ 1 &= 26 - 5 \times 5 \\ 1 &= 1(26) + (-5)(5) \end{aligned}$$

The coefficient of 5 is -5. We convert this negative result to a positive value within the modulus:

$$y \equiv -5 + 26 \equiv \mathbf{21} \pmod{\mathbf{26}} \tag{2}$$

Affine Cipher-Exercise

Values

$$a = 5, b = 8,$$

$$a^{-1} = 21 mod 26.$$

Affine Cipher-Exercise

Values

$$a = 5, b = 8,$$

$$a^{-1} = 21 mod 26.$$

Recovered Plaintext

Cryptography is the science of securing information

Affine Cipher-Exercise

Values

$$a = 5, b = 8,$$

$$a^{-1} = 21 mod 26.$$

Recovered Plaintext

Cryptography is the science of securing information by transforming it into a form that is difficult to understand without the proper key. Classical ciphers such as the Affine cipher provide a simple introduction to the concepts of encryption and decryption.

Scytale Cipher

Plaintext

HELLOWORLD

Scytale Cipher

Key: Using a scytale with 3 rows:

Scytale Cipher

Key: Using a scytale with 3 rows:

Recovered Plaintext

HOLEWDLOLR

Scytale Cipher

Key: Using a scytale with 3 rows:

Recovered Plaintext

HOLEWDLOLR

Key: Using a scytale with 4 rows:

Scytale Cipher

Key: Using a scytale with 3 rows:

Recovered Plaintext

HOLEWDLOLR

Key: Using a scytale with 4 rows:

Recovered Plaintext

HLODEORLWL

Rail Fence Cipher

Plaintext

HELLOWORLD

Rail Fence Cipher

Key: Using a fence with 3 rows:

Rail Fence Cipher

Key: Using a fence with 3 rows:

Recovered Plaintext

HOLELWRDLO

Key: Using a fence with 4 rows:

Rail Fence Cipher

Key: Using a fence with 3 rows:

Recovered Plaintext

HOLELWRDLO

Key: Using a fence with 4 rows:

Recovered Plaintext

HOEWRLLLDOO

Outline

1. Substitution Ciphers
   - Caesar Cipher
   - Shift Cipher
   - Atbash Cipher
   - Affine Cipher
2. Transposition Ciphers
   - Scytale Cipher
   - Rail Fence Cipher
3. Polygraphic Ciphers
   - Hill Cipher
4. Polyalphabetic Ciphers
   - Vigenère Cipher

Hill Cipher

Plaintext

HELP

Hill Cipher

Plaintext

HELP

Key Matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Hill Cipher

Plaintext

HELP

Key Matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Digraph	Plaintext Vector	Cipher Vector (KP mod 26)	Ciphertext
HE	$\begin{bmatrix} 7 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$	HI

Hill Cipher

Plaintext

HELP

Key Matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Digraph	Plaintext Vector	Cipher Vector (KP mod 26)	Ciphertext
HE	$\begin{bmatrix} 7 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$	HI
LP	$\begin{bmatrix} 11 \\ 15 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 19 \end{bmatrix}$	AT

Ciphertext

HIAT

Outline

1. Substitution Ciphers
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3. Polygraphic Ciphers
   - Hill Cipher
4. Polyalphabetic Ciphers
   - Vigenère Cipher

Vigenère Cipher

Plaintext

ATTACKATDAWN

Vigenère Cipher

Plaintext

ATTACKATDAWN

Key

LEMON

Vigenère Cipher

Plaintext

ATTACKATDAWN

Key

LEMON

Ciphertext

LXFOPVEFRNHR

Vigenère Cipher

Plaintext	Key	Calculation $(P + K) \bmod 26$	Ciphertext
A	L	$(0 + 11) \bmod 26 = 11$	L

Vigenère Cipher

Plaintext	Key	Calculation $(P + K)$ mod 26	Ciphertext
A	L	$(0 + 11) \bmod 26 = 11$	L
T	E	$(19 + 4) \bmod 26 = 23$	X
T	M	$(19 + 12) \bmod 26 = 5$	F
A	O	$(0 + 14) \bmod 26 = 14$	O
C	N	$(2 + 13) \bmod 26 = 15$	P
K	L	$(10 + 11) \bmod 26 = 21$	V

Vigenère Cipher

Plaintext	Key	Calculation $(P+K)$ mod 26	Ciphertext
A	L	$(0+11) \bmod 26 = 11$	L
T	E	$(19+4) \bmod 26 = 23$	X
T	M	$(19+12) \bmod 26 = 5$	F
A	O	$(0+14) \bmod 26 = 14$	O
C	N	$(2+13) \bmod 26 = 15$	P
K	L	$(10+11) \bmod 26 = 21$	V
A	E	$(0+4) \bmod 26 = 4$	E
T	M	$(19+12) \bmod 26 = 5$	F
D	O	$(3+14) \bmod 26 = 17$	R
A	N	$(0+13) \bmod 26 = 13$	N
W	L	$(22+11) \bmod 26 = 7$	H
N	E	$(13+4) \bmod 26 = 17$	R

Vigenère Cipher Cryptanalysis-Excercise

Ciphertext

CHREEVOAHMAERATBIAXXWTNXBEEOPHBSBQMQEQER
BWRVXUOAKXAOSXXWEAHBWGJMMQMNKGRFVGXWTRZX
WIAKLXFPSKAUTEMNDCMGTSXMXBTUIADNGMGPSREL
XNJELXVRVPRTULHDNQWTWDTYGBPHXTFALJHASVBF
XNGLLCHRZBWELEKMSJIKNBHWRJGNMGJSGLXFEYPH
AGNRBIEQJTAMRVLCRREMNDGLXRRIMGNSNRWCHRQH
AEYEVTAQEBBIPEEWEVKAKOEWADREMXMTBHHCHRTK
DNVRZCHRCLQOHPWQAIIWXNRMGWOIIFKEE

Vigenère Cipher Cryptanalysis-Excercise

Ciphertext

CHREEVOAHMAERATBIAXXWTNXBEEOPHBSBQMQEQER
BWRVXUOAKXAOSXXWEAHBWGJMMQMNKGRFVGXWTRZX
WIAKLXFPSKAUTEMNDCMGTSXMXBTUIADNGMGPSREL
XNJELXVRVPRTULHDNQWTWDTYGBPHXTFALJHASVBF
XNGLLCHRZBWELEKMSJIKNBHWRJGNMGJSGLXFEYPH
AGNRBIEQJTAMRVLCRREMNDGLXRRIMGNSNRWCHRQH
AEYEVTAQEBBIPEEWEVKAKOEWADREMXMTBHHCHRTK
DNVRZCHRCLQOHPWQAIIWXNRMGWOIIFKEE

Methods for estimating key length:

- Kasiski Examination
- Index of Coincidence

Vigenère Cipher Cryptanalysis-Excercise

Figure: Ciphertext

Kasiski Examination

First, let us apply the Kasiski test. The ciphertext string **CHR** occurs five times in the ciphertext, beginning at positions

Kasiski Examination

First, let us apply the Kasiski test. The ciphertext string **CHR** occurs five times in the ciphertext, beginning at positions

$$1,\ 166,\ 236,\ 276,\ \text{and } 286.$$

The distances from the first occurrence to the remaining occurrences are

$$166-1=165,\ 236-1=235,\ 276-1=275,\ 286-1=285.$$

Thus, the distances are

$$165,\ 235,\ 275,\ 285.$$

Computing their greatest common divisor (GCD),

$$\gcd(165,235,275,285)=5.$$

Therefore, the most likely keyword length is

$$m=5.$$

Index of Coincidence

$$IC(x) = \frac{\sum_{i=0}^{25} \binom{f_i}{2}}{\binom{n}{2}} = \frac{\sum_{i=0}^{25} f_i(f_i - 1)}{n(n-1)}.$$

Notation:

- n: Total number of letters in the text.
- f_i: Frequency of the i-th letter of the alphabet.
- $\sum_{i=0}^{25}$: Sum over all 26 letters (A–Z).

Type	IC
English	0.066
Random	0.038
Vigenère	0.038–0.066

Index of Coincidence

Next, we verify this result using the Index of Coincidence (IC).

- For $m = 1$, the index of coincidence is:
$$0.045.$$
- For $m = 2$, the two indices are:
$$0.046, \; 0.041.$$
- For $m = 3$, the three indices are:
$$0.043, \; 0.050, \; 0.047.$$
- For $m = 4$, the four indices are:
$$0.042, \; 0.039, \; 0.045, \; 0.040.$$
- For $m = 5$, the five indices are:
$$0.063, \; 0.068, \; 0.069, \; 0.061, \; 0.072.$$

Index of Coincidence

Since the indices obtained for $m = 5$ are significantly closer to the expected Index of Coincidence for English text (approximately 0.065), this provides strong evidence that the

$$\boxed{\text{keyword length is } \mathbf{5}}.$$

Thus, both the Kasiski examination and the Index of Coincidence method indicate that the Vigenère cipher keyword has length 5.

Suppose that $0 \leq g \leq 25,$ and define the quantity

$$M_g = \sum_{i=0}^{25} \frac{p_i f_{(i+g)}}{n}.$$

Key Recovery

i	value of $M_g(\mathbf{y}_i)$								
1	.035	.031	.036	.037	.035	.039	.028	.028	.048
	$\boxed{.061}$	.039	.032	.040	.038	.038	.045	.036	.030
	.042	.043	.036	.033	.049	.043	.042	.036	
2	$\boxed{.069}$	.044	.032	.035	.044	.034	.036	.033	.029
	.031	.042	.045	.040	.045	.046	.042	.037	.032
	.034	.037	.032	.034	.043	.032	.026	.047	
3	.048	.029	.042	.043	.044	.034	.038	.035	.032
	.049	.035	.031	.035	$\boxed{.066}$	.035	.038	.036	.045
	.027	.035	.034	.034	.036	.035	.046	.040	
4	.045	.032	.033	.038	$\boxed{.060}$	.034	.034	.034	.050
	.033	.033	.043	.040	.033	.029	.036	.040	.044
	.037	.050	.034	.034	.039	.044	.038	.035	
5	.034	.031	.035	.044	.047	.037	.043	.038	.042
	.037	.033	.032	.036	.037	.036	.045	.032	.029
	.044	$\boxed{.072}$	.037	.027	.031	.048	.036	.037	

From the table, we see that the key is likely to be K $= (9, 0, 13, 4, 19)$,

Key Recovery

From the table, we see that the key is likely to be $K = (9, 0, 13, 4, 19)$. Hence the keyword likely is **JANET**. This is correct, and the complete decryption of the ciphertext is the following:

Recovered Plaintext

The almond tree was in tentative blossom.

Key Recovery

From the table, we see that the key is likely to be $K = (9, 0, 13, 4, 19)$. Hence the keyword likely is **JANET**. This is correct, and the complete decryption of the ciphertext is the following:

Recovered Plaintext

The almond tree was in tentative blossom. The days were longer, often ending with magnificent evenings of corrugated pink skies. The hunting season was over, with hounds and guns put away for six months. The vineyards were busy again as the well-organized farmers treated their vines and the more lackadaisical neighbors hurried to do the pruning they should have done in November.

Questions?